Pre-requisites

None — start from zero

A laptop, an internet connection and 2 hours a day

How to read this roadmap

- Main subject (hours · days at 2 h/day)
- Essential topic
- Recommended topic
- Optional topic
- 1 Learning order inside a subject

AI/ML Engineer

From scratch to advanced

Total duration

2,755 hours

10 phases · 55 subjects · 482 topics

≈ 63 months at 2 h/day

Phases

0. Computing Foundations	82 h
1. Programming & Software Engineering	265 h
2. Mathematics for AI (parallel track)	215 h
3. DSA, Data & Systems	370 h
4. Classical Machine Learning	285 h
5. Deep Learning	343 h
6. Generative AI & LLMs	285 h
7. Production AI Engineering	460 h
8. Advanced & Research	290 h
9. Capstone & Job Readiness	160 h

PHASE 0

Computing Foundations

82 hours · ≈ 41 study days at 2 h/day

S1 · Computing Fundamentals

15 h · ≈ 8 days at 2 h/day

1 Data Representation

Binary & Hex · Bits & Bytes · Integers (Two's Complement) · Floating Point (IEEE 754) · float32 / float16 / bfloat16 · ASCII, Unicode, UTF-8

4 h

2 Hardware Basics

CPU & Cores · Cache (L1/L2/L3) · RAM vs Disk vs SSD · Latency Numbers · What a GPU Is

4 h

3 How Programs Run

Compiler vs Interpreter · Processes vs Threads · Stack vs Heap · Python Interpreter

4 h

4 Operating System Concepts

Scheduling · Virtual Memory · File Systems · System Calls · Permissions

Recommended 2 h

5 Computer Architecture

Instruction Pipelines · SIMD / Vector Instructions

Optional 1 h

S2 · Linux & the Shell

35 h · ≈ 18 days at 2 h/day

1 Navigation & Files

pwd, ls, cd · mkdir, cp, mv, rm · find · Absolute vs Relative Paths · Filesystem Layout

5 h

2 Viewing & Editing

cat, less, head, tail -f · nano / vim Basics · grep, wc, sort, uniq, cut · Pipes · Redirection

5 h

3 Permissions & Users

chmod, chown · sudo · Permission Bits · Groups

4 h

4 Processes

ps, top, htop · kill & Signals · Background Jobs · Exit Codes

4 h

5 Packages & Environment

apt · Environment Variables · PATH · .bashrc

4 h

6 Networking from Shell

curl, wget · ssh, scp · SSH Keys · ping · Ports

4 h

7 Shell Scripting

Variables · Conditionals · Loops · Functions · Arguments · set -euo pipefail · cron

4 h

8 Text Processing

sed · awk · jq · xargs

Recommended 2 h

9 Remote Work

tmux / screen · rsync · df, du · nvidia-smi

Recommended 2 h

10 Advanced Linux

systemd · journalctl · cgroups & namespaces

Optional 1 h

S3 · Git & GitHub

22 h · ≈ 11 days at 2 h/day

1 Core Model

Repository · Working Tree · Staging Area · Commits & HEAD · add, commit, status, log, diff

3 h

2 Branching

branch, switch · merge · Merge Conflicts

3 h

3 Remotes

clone, push, pull, fetch · origin & upstream · SSH Auth

3 h

4 Collaboration

Forks · Pull Requests · Code Review · Issues · Commit Messages

3 h

5 Hygiene

.gitignore · Keeping Secrets Out · Large Files

3 h

6 History Editing

rebase · Interactive Rebase · cherry-pick · reset vs revert · stash · reflog

Recommended 2 h

7 Workflows

Trunk-Based · GitFlow · Tags & Releases · Semantic Versioning

Recommended 2 h

8 Large Files

Git LFS · DVC / HF Hub

Recommended 2 h

9 Git Internals

Blob, Tree, Commit Objects · Snapshots

Optional 1 h

S4 · Development Environment

10 h · ≈ 5 days at 2 h/day

1 VS Code

Extensions · Debugger · Integrated Terminal · Remote-SSH / WSL

1 h

2 Python Environments

Virtual Environments · venv & pip · requirements.txt · uv

2 h

3 Jupyter Notebooks

Cells & Kernels · Magic Commands · Notebooks vs Scripts

2 h

4 Free GPU Platforms

Google Colab · Kaggle Notebooks

2 h

5 Conda / Mamba

Non-Python Dependencies · CUDA Toolkits

Recommended 1 h

6 Linters & Formatters

Ruff · Black · pre-commit

Recommended 1 h

7 Dev Containers

devcontainer.json

Optional 1 h

Continues: Phase 1 · Programming & Software Engineering

PHASE 1

Programming & Software Engineering

265 hours · ≈ 133 study days at 2 h/day

S5 — Python — Beginner

85 h · ≈ 43 days at 2 h/day

1 Basics

Variables · Types · Operators · Input / Output · f-strings · Comments

10 h

2 Control Flow

if / elif / else · for & while · break / continue · range, enumerate, zip

10 h

3 Data Structures

Lists · Tuples · Dictionaries · Sets · Indexing & Slicing · Mutability · Shallow vs Deep Copy

10 h

4 Functions

Parameters & Returns · Default & Keyword Args · *args / **kwargs · Scope (LEGB) · Docstrings

10 h

5 Strings

String Methods · Formatting · Split & Join · Regular Expressions

9 h

6 Files

Text Files · CSV · JSON · with Statement · pathlib

9 h

7 Errors

Reading Tracebacks · try / except / finally · Raising Exceptions

9 h

8 Modules

import · Standard Library · pip / uv Install

9 h

9 Comprehensions

List / Dict / Set · lambda · map & filter · sorted with key

9 h

S6 — Python — Intermediate & OOP

70 h · ≈ 35 days at 2 h/day

1 Classes & Objects

__init__ · Attributes & Methods · self · Class vs Instance Attributes

9 h

2 OOP Pillars

Encapsulation · Inheritance · Polymorphism · Abstraction · super() · Abstract Base Classes

9 h

3 Dunder Methods

__repr__ / __str__ · __len__ / __getitem__ · __iter__ · __call__ · Operator Overloading

8 h

4 Composition vs Inheritance

When to Use Each

8 h

5 Dataclasses & Type Hints

dataclasses · typing Module · Optional / Union · Generics · mypy / pyright

8 h

6 Iterators & Generators

Iterator Protocol · yield · Generator Expressions · Lazy Evaluation

8 h

7 Project Structure

Packages · __init__.py · Imports · __main__ · argparse / Typer

8 h

8 Logging

logging Module · Log Levels · Handlers

8 h

9 Functional Tools

functools · lru_cache · partial · itertools

Recommended · 4 h

S7 — Python — Advanced

65 h · ≈ 33 days at 2 h/day

1 Decorators

Functions as Objects · Closures · Decorators with Args · functools.wraps

10 h

2 Context Managers

__enter__ / __exit__ · contextlib

9 h

3 Concurrency

The GIL · threading · multiprocessing · concurrent.futures · I/O vs CPU Bound

9 h

4 asyncio

async / await · Event Loop · gather · Semaphores & Rate Limits

9 h

5 Packaging

pyproject.toml · Wheels · Versioning · Publishing to PyPI · Lock Files

9 h

6 Performance

cProfile · line_profiler · py-spy · Memory Profiling · Vectorizing

9 h

7 Advanced Typing

Protocol · TypedDict · Generics · Pydantic

Recommended · 4 h

8 Data Model

Descriptors · __slots__ · Metaclasses

Recommended · 4 h

9 Extending Python

Cython · Numba · Rust (PyO3) · Free-threaded Python

Optional · 2 h

S8 — Software Engineering & Testing

45 h · ≈ 23 days at 2 h/day

1 Clean Code

Naming · Small Functions · Single Responsibility · DRY · Comments that Explain Why

6 h

2 Design Principles

SOLID · Coupling vs Cohesion

7 h

3 Testing

pytest · Fixtures · parametrize · Mocking · Unit / Integration / E2E · Coverage · Testing ML Code

7 h

4 Debugging

VS Code Debugger · pdb · Bisecting

7 h

5 Code Review

Giving & Receiving · Small PRs

7 h

6 Design Patterns

Factory · Strategy · Observer · Adapter · Dependency Injection

Recommended · 3 h

7 Documentation

READMEs · Docstrings · ADRs · MkDocs

Recommended · 3 h

8 Config & Errors

Fail Fast · YAML / Hydra · 12-Factor App

Recommended · 3 h

9 Formal Design

UML · Domain-Driven Design

Optional · 2 h

Continues: Phase 2 · Mathematics for AI (parallel track)

PHASE 2

Mathematics for AI (parallel track)

215 hours · ≈ 108 study days at 2 h/day

S9 Linear Algebra

70 h · ≈ 35 days at 2 h/day

1 Vectors
- Geometric Meaning
- Addition & Scaling
- Dot Product
- Norms (L1, L2)
- Cosine Similarity
- Projections

9 h

2 Matrices
- Linear Transformations
- Matrix Multiply
- Shapes & Broadcasting
- Transpose
- Identity & Inverse

9 h

3 Linear Systems
- Gaussian Elimination
- Rank
- Linear Independence
- Span & Basis
- Column & Null Space

9 h

4 Determinants
- Volume Meaning
- Invertibility

9 h

5 Eigenvalues & Eigenvectors
- Computing Them
- Diagonalization
- Link to PCA

8 h

6 SVD
- Decomposition
- Low-Rank Approximation
- PCA via SVD

8 h

7 Orthogonality
- Orthonormal Matrices
- Gram–Schmidt
- QR

8 h

8 Positive Definite Matrices
- Quadratic Forms
- Covariance Matrices

Recommended 4 h

9 Tensors
- N-D Arrays
- einsum
- Batch / Sequence / Head Shapes

Recommended 4 h

10 Numerical Linear Algebra
- Conditioning
- Stability

Optional 2 h

S10 Calculus & Optimization

60 h · ≈ 30 days at 2 h/day

1 Derivatives
- Limits
- Derivative Rules
- exp, log, sigmoid

8 h

2 Chain Rule
- Composite Functions
- Basis of Backprop

7 h

3 Partial Derivatives & Gradients
- Multivariable Functions
- Gradient Direction

7 h

4 Vector & Matrix Calculus
- Jacobian
- Gradient of Matmul
- Hessian

7 h

5 Automatic Differentiation
- Computational Graphs
- Forward vs Reverse Mode

7 h

6 Optimization
- Gradient Descent
- Learning Rate
- Local vs Global Minima
- Saddle Points
- Convexity

7 h

7 Gradient Descent Variants
- SGD
- Mini-batch
- Momentum
- RMSProp
- Adam

7 h

8 Integrals
- Area Under Curve
- Expectations
- AUC

Recommended 4 h

9 Taylor Series
- Local Approximation
- Second-Order Methods

Recommended 4 h

10 Constrained Optimization
- Lagrange Multipliers
- KKT Conditions

Optional 2 h

S11 Probability & Statistics

85 h · ≈ 43 days at 2 h/day

1 Descriptive Statistics
- Mean / Median / Mode
- Variance & Std Dev
- Percentiles
- Skew
- Outliers

9 h

2 Probability Basics
- Sample Spaces
- Events
- Conditional Probability
- Independence

9 h

3 Bayes' Theorem
- Prior
- Likelihood
- Posterior
- Naive Bayes

9 h

4 Distributions
- Bernoulli
- Binomial
- Categorical
- Poisson
- Normal
- Exponential
- PMF / PDF / CDF

8 h

5 Expectation & Covariance
- Expectation
- Variance
- Covariance
- Correlation vs Causation
- Multivariate Gaussian

8 h

6 Limit Theorems
- Law of Large Numbers
- Central Limit Theorem

8 h

7 Estimation
- MLE
- MAP
- Cross-Entropy = NLL

8 h

8 Inferential Statistics
- Sampling
- Confidence Intervals
- Hypothesis Tests
- p-values
- t-test
- Chi-square

8 h

9 Experiment Design
- A/B Testing
- Statistical Power
- Sample Size

8 h

10 Information Theory
- Entropy
- Cross-Entropy
- KL Divergence
- Perplexity
- Mutual Information

Recommended 4 h

11 Resampling
- Bootstrap
- Bayesian Inference

Recommended 4 h

12 Stochastic Processes
- Markov Chains
- Monte Carlo

Optional 2 h

Continues: Phase 3 · DSA, Data & Systems

PHASE 3

DSA, Data & Systems

370 hours · ≈ 185 study days at 2 h/day

S12 · Data Structures & Algorithms

110 h · ≈ 55 days at 2 h/day

1 · Complexity — 11 h

- Big-O Time & Space
- Best / Average / Worst
- Amortized

2 · Arrays & Strings — 11 h

- Two Pointers
- Sliding Window
- Prefix Sums

3 · Hashing — 10 h

- Hash Maps
- Hash Sets
- Collisions
- Frequency Counting

4 · Linear Structures — 10 h

- Linked Lists
- Stacks
- Queues
- Deques

5 · Recursion — 10 h

- Recursion
- Backtracking

6 · Sorting & Searching — 10 h

- Merge Sort
- Quicksort
- Binary Search
- Search on Answer

7 · Trees — 10 h

- Binary Trees
- Traversals
- BST
- Tries

8 · Heaps — 10 h

- Priority Queues
- Top-K Problems

9 · Graphs — 10 h

- Representations
- BFS
- DFS
- Topological Sort
- Dijkstra
- Union-Find

10 · Dynamic Programming — 10 h

- Memoization
- Tabulation
- Knapsack
- LCS
- Edit Distance

11 · More Patterns — 5 h

Recommended

- Greedy
- Intervals
- Bit Manipulation

12 · Advanced Structures — 3 h

Optional

- Segment Trees
- HNSW Graphs
- k-d Trees

S13 · Data Analysis

70 h · ≈ 35 days at 2 h/day

1 · NumPy — 11 h

- ndarrays
- dtypes
- Indexing & Masks
- Broadcasting
- Vectorization
- Axis Ops
- Random

2 · pandas — 11 h

- Series & DataFrame
- Read CSV / Parquet / SQL
- loc / iloc
- Filtering
- groupby
- merge / join
- pivot / melt
- Datetime
- Categoricals

3 · Data Cleaning — 11 h

- Missing Values
- Duplicates
- Type Fixes
- Outliers
- String Cleanup

4 · Visualization — 11 h

- Matplotlib
- Seaborn
- Histograms
- Box Plots
- Scatter
- Heatmaps
- Choosing Charts

5 · EDA Workflow — 11 h

- Questions
- Summary Stats
- Distributions
- Relationships
- Data Quality
- Write-up

6 · Interactive Viz — 6 h

Recommended

- Plotly
- Streamlit
- Gradio

7 · Scaling Up — 6 h

Recommended

- Polars
- DuckDB
- Chunked Processing

8 · Big Data — 3 h

Optional

- PySpark

S14 · SQL & Relational Databases

45 h · ≈ 23 days at 2 h/day

1 · Relational Model — 5 h

- Tables
- Primary & Foreign Keys
- Data Types

2 · Querying — 5 h

- SELECT / WHERE
- ORDER BY / LIMIT
- DISTINCT
- Aggregates
- GROUP BY / HAVING

3 · Joins — 5 h

- Inner
- Left / Right
- Full
- Self Join

4 · Subqueries & CTEs — 5 h

- Subqueries
- WITH Clauses

5 · Window Functions — 5 h

- ROW_NUMBER
- RANK
- LAG / LEAD
- Running Totals

6 · Writing Data — 4 h

- INSERT / UPDATE / DELETE
- Transactions
- ACID

7 · Schema Design — 4 h

- Normalization
- Denormalization
- Star Schema

8 · Performance — 4 h

- B-tree Indexes
- EXPLAIN ANALYZE
- Query Plans

9 · SQL from Python — 4 h

- psycopg
- SQLAlchemy
- pandas.read_sql

10 · Data Warehouses — 2 h

Recommended

- Columnar Storage
- BigQuery
- Snowflake

11 · pgvector — 2 h

Recommended

- Vector Search in Postgres

S15 · NoSQL Databases

20 h · ≈ 10 days at 2 h/day

1 · Why NoSQL — 6 h

- CAP Theorem
- Schema Flexibility
- Horizontal Scaling

2 · Document Stores — 5 h

- MongoDB
- CRUD
- Indexes
- Aggregation

3 · Key-Value Stores — 5 h

- Redis
- TTL
- Caching Patterns
- Rate Limiting
- Pub/Sub
- Queues

4 · Choosing a Database — 3 h

Recommended

- SQL vs Document vs KV vs Vector vs Graph

5 · Other Types — 1 h

Optional

- Cassandra
- Neo4j

S16 · Data Engineering Basics

40 h · ≈ 20 days at 2 h/day

1 · Pipelines — 6 h

- ETL vs ELT
- Batch vs Streaming

2 · File Formats — 6 h

- CSV
- JSON / JSONL
- Parquet
- Arrow
- Partitioning

3 · Data Quality — 6 h

- Pydantic
- Pandera
- Great Expectations
- Data Contracts

4 · Storage — 6 h

- Data Lakes
- Warehouses
- Lakehouses

5 · Data for LLMs — 6 h

- Scraping Ethics
- Document Parsing
- Deduplication
- PII Removal

6 · Orchestration — 3 h

Recommended

- DAGs
- Airflow
- Prefect
- Dagster

7 · Transformation — 3 h

Recommended

- dbt

8 · Streaming — 3 h

Recommended

- Kafka Topics
- Partitions
- Consumers

9 · Distributed Processing — 1 h

Optional

- Spark
- Ray Data

S17 · Computer Networks

25 h · ≈ 13 days at 2 h/day

1 · Network Models — 3 h

- OSI Layers
- TCP/IP

2 · Addressing — 3 h

- IPv4 / IPv6
- Subnets & CIDR
- Ports
- NAT

3 · Transport — 4 h

- TCP vs UDP
- Handshake
- Sockets

4 · DNS — 4 h

- Resolution
- Record Types
- Caching

5 · HTTP / HTTPS — 4 h

- Methods
- Status Codes
- Headers
- HTTP/2 & 3
- TLS

6 · Traffic — 4 h

- Load Balancers
- Reverse Proxy
- CDN
- Firewalls

7 · Real-time Protocols — 2 h

Recommended

- WebSockets
- Server-Sent Events
- gRPC

8 · Interconnects — 1 h

Optional

- InfiniBand
- RDMA
- NVLink

S18 · APIs & Backend (FastAPI)

60 h · ≈ 30 days at 2 h/day

1 · API Design — 7 h

- REST
- Resources & Verbs
- Status Codes
- Versioning
- Pagination
- Idempotency

2 · FastAPI — 7 h

- Routes
- Path / Query / Body
- Pydantic Models
- OpenAPI Docs
- Dependencies

3 · Async & Streaming — 7 h

- Async Endpoints
- Background Tasks
- SSE Streaming

4 · Databases — 7 h

- SQLAlchemy ORM
- Alembic Migrations
- Connection Pooling

5 · Auth — 7 h

- API Keys
- OAuth2
- JWT
- Password Hashing
- RBAC

6 · Testing APIs — 7 h

- TestClient
- httpx
- Test Database

7 · Production Basics — 7 h

- Validation
- Error Handling
- Logging
- CORS
- Rate Limiting

8 · Other Frameworks — 3 h

Recommended

- Flask
- Django
- gRPC

9 · Demo Frontends — 3 h

Recommended

- Streamlit
- Gradio
- Basic HTML / JS

9 · Task Queues — 3 h

Recommended

- Celery
- RQ
- Arq

11 · Other APIs — 2 h

Optional

- GraphQL
- WebSockets

Continues: Phase 4 · Classical Machine Learning

PHASE 4

Classical Machine Learning

285 hours · ≈ 143 study days at 2 h/day

S19 ML Foundations

30 h · ≈ 15 days at 2 h/day

1 What ML Is

Rules vs Learning | Supervised / Unsupervised / RL | When Not to Use ML

4 h

2 ML Workflow

Frame | Collect | Split | Baseline | Train | Evaluate | Deploy | Monitor

4 h

3 Data Splits

Train / Val / Test | Data Leakage

4 h

4 Loss & Optimization

Loss Functions | Empirical Risk | Gradient Descent in NumPy

4 h

5 Generalization

Underfitting | Overfitting | Bias–Variance | Learning Curves

4 h

6 Regularization

L1 / Lasso | L2 / Ridge | Early Stopping

4 h

7 Baselines

Majority Class | Heuristics

4 h

8 Learning Theory

No Free Lunch | Curse of Dimensionality | Inductive Bias

Recommended 2 h

S20 Supervised Learning

90 h · ≈ 45 days at 2 h/day

1 Linear Regression

Normal Equation | Gradient Descent | Assumptions | Polynomial Features

9 h

2 Logistic Regression

Sigmoid | Log-loss | Decision Boundary | Softmax

9 h

3 k-Nearest Neighbours

Distance Metrics | Choosing k | Scaling

10 h

4 Naive Bayes

Text Classification

10 h

5 Decision Trees

Gini / Entropy | Depth | Pruning

10 h

6 Ensembles

Bagging | Random Forests | Boosting | XGBoost | LightGBM | CatBoost

10 h

7 SVM

Margins | Hinge Loss | Kernels

10 h

8 Imbalanced Data

Class Weights | SMOTE | Threshold Tuning

10 h

9 Time-Series Forecasting

Lag Features | Rolling Features | Time Splits | ARIMA / Prophet

Recommended 5 h

10 Ranking & Recommendation

Learning to Rank | Collaborative Filtering | Matrix Factorization

Recommended 5 h

11 Other Models

GLMs | Gaussian Processes

Optional 2 h

S21 Unsupervised Learning

45 h · ≈ 23 days at 2 h/day

1 Clustering

k-means / k-means++ | Elbow & Silhouette | Hierarchical | DBSCAN / HDBSCAN

10 h

2 Mixture Models

Gaussian Mixtures | EM Algorithm

9 h

3 Dimensionality Reduction

PCA | Explained Variance | t-SNE | UMAP

9 h

4 Anomaly Detection

Isolation Forest | One-class SVM | Statistical Methods

9 h

5 Association Rules

Apriori | Market Basket

Recommended 4 h

6 Topic Modelling

LDA | BERTopic

Recommended 4 h

S22 Semi- & Self-Supervised Learning

15 h · ≈ 8 days at 2 h/day

1 Problem Setting

Labelled vs Unlabelled | Cluster Assumption

3 h

2 Classical Methods

Self-Training | Pseudo-Labels | Label Propagation

4 h

3 Self-Supervised Learning

Pretext Tasks | Contrastive (SimCLR, CLIP) | Masked Modelling | Next-Token Prediction

4 h

4 Consistency Methods

MixMatch | FixMatch

Recommended 2 h

5 Label Efficiency

Active Learning | Weak Supervision | LLM Labelling

Recommended 2 h

S23 Feature Engineering

30 h · ≈ 15 days at 2 h/day

1 Numeric Features

Standard / Min-Max / Robust Scaling | Log Transform | Binning

3 h

2 Categorical Features

One-Hot | Ordinal | Target Encoding | Hashing

3 h

3 Missing Data

Imputation | Missing Indicators

4 h

4 Datetime Features

Cyclical Encoding | Lags | Rolling Windows | Holidays

4 h

5 Text Features

Bag of Words | TF-IDF | n-grams

4 h

6 Feature Selection

Correlation | Mutual Information | L1 | Permutation Importance | RFE

4 h

7 Pipelines

sklearn Pipeline | ColumnTransformer

4 h

8 Interactions

Feature Crosses | Domain Features

Recommended 2 h

9 Feature Stores

Offline / Online | Training–Serving Skew

Recommended 2 h

S24 Model Evaluation & Optimization

35 h · ≈ 18 days at 2 h/day

1 Classification Metrics

Confusion Matrix | Precision / Recall / F1 | ROC-AUC | PR-AUC | Log-loss | Thresholds

3 h

2 Regression Metrics

MAE | MSE / RMSE | R^2 | MAPE

4 h

3 Ranking Metrics

Precision@k | Recall@k | MRR | NDCG

4 h

4 Cross-Validation

k-Fold | Stratified | Group | Time-Series Split

4 h

5 Hyperparameter Tuning

Grid Search | Random Search | Bayesian (Optuna)

4 h

6 Error Analysis

Slice Metrics | Worst Errors

4 h

7 Calibration

Reliability Curves | Platt Scaling | Isotonic

4 h

8 Interpretability

Feature Importance | SHAP | Partial Dependence

4 h

9 Model Comparison

Confidence Intervals | Bootstrap | McNemar's Test

Recommended 2 h

10 Fairness Metrics

Group Metrics

Recommended 2 h

S25 Classical ML Capstone Project

40 h · ≈ 20 days at 2 h/day

1 Problem & Data

Problem Framing | Dataset | EDA

10 h

2 Modelling

Features | 3+ Models | Tuning

10 h

3 Evaluation

Honest Test | SHAP | Error Analysis

10 h

4 Delivery

Prediction API | Model Card | Report

10 h

Continues: Phase 5 · Deep Learning

PHASE 5

Deep Learning

343 hours ÷ ≈ 172 study days at 2 h/day

S26 Neural Network Foundations

50 h ÷ ≈ 25 days at 2 h/day

1 Neurons & MLPs
- Perceptron
- Multi-Layer Perceptron
- Universal Approximation

6 h

2 Activations
- Sigmoid
- tanh
- ReLU
- Leaky ReLU
- GELU
- SiLU
- Softmax

6 h

3 Forward Pass & Loss
- MSE
- Cross-Entropy
- Binary Cross-Entropy

5 h

4 Backpropagation
- Computational Graphs
- micrograd from Scratch

5 h

5 Optimizers
- SGD
- Momentum
- Nesterov
- RMSProp
- Adam / AdamW
- LR Warmup
- Cosine Decay

5 h

6 Initialization
- Xavier / Glorot
- He / Kaiming

5 h

7 Normalization
- BatchNorm
- LayerNorm
- RMSNorm

5 h

8 Regularization
- Dropout
- Weight Decay
- Augmentation
- Early Stopping

5 h

9 Training Diagnostics
- Loss Curves
- Gradient Norms
- Overfit One Batch
- Vanishing / Exploding
- Residuals

5 h

10 Mixed Precision
- fp16 / bf16
- Gradient Clipping

Recommended 3 h

S27 PyTorch

50 h ÷ ≈ 25 days at 2 h/day

1 Tensors
- Creation
- dtypes
- Devices
- Indexing
- view / reshape / permute
- einsum

5 h

2 Autograd
- requires_grad
- backward
- no_grad
- detach

5 h

3 nn.Module
- Layers
- Parameters
- Custom Models
- Sequential

6 h

4 Data Loading
- Dataset
- DataLoader
- Batching
- num_workers
- collate_fn
- Transforms

6 h

5 Training Loop
- Forward
- Loss
- Backward
- Step
- zero_grad
- train() / eval()

6 h

6 Checkpoints
- state_dict
- Resume Training
- safetensors

6 h

7 GPU Usage
- .to(device)
- Memory Management
- CUDA Memory Stats

6 h

8 High-Level Libraries
- PyTorch Lightning
- HF Accelerate

Recommended 3 h

9 Performance
- torch.compile
- torch.amp
- Profiler

Recommended 3 h

10 TensorFlow / Keras
- Reading TF Code
- Keras 3

Recommended 3 h

11 JAX
- jit
- grad
- vmap

Optional 1 h

S28 CNNs & Computer Vision

65 h ÷ ≈ 33 days at 2 h/day

1 Image Basics
- Pixels & Channels
- Colour Spaces
- OpenCV / PIL
- Resize & Normalize

7 h

2 Convolutions
- Kernels
- Stride & Padding
- Receptive Field
- Pooling
- Weight Sharing

8 h

3 Architectures
- LeNet
- AlexNet
- VGG
- ResNet
- Inception
- EfficientNet
- MobileNet

8 h

4 Transfer Learning
- Pretrained Models
- Fine-tuning
- torchvision
- timm

8 h

5 Augmentation
- Flips & Crops
- Colour Jitter
- Mixup / CutMix
- Albumentations

8 h

6 Object Detection
- Bounding Boxes
- IoU
- Anchors
- NMS
- Faster R-CNN
- YOLO
- mAP

8 h

7 Segmentation
- Semantic
- Instance
- Panoptic
- U-Net
- Mask R-CNN
- SAM

8 h

8 Vision Transformers
- ViT
- CLIP

Recommended 4 h

9 Documents & Video
- OCR
- Document AI
- Frames & Tracking

Recommended 4 h

10 3D & Pose
- 3D Vision
- Pose Estimation
- Depth Estimation

Optional 2 h

S29 RNNs, LSTMs & GRUs

28 h ÷ ≈ 14 days at 2 h/day

1 Sequence Modelling
- Variable Length
- Teacher Forcing
- Many-to-One / Many-to-Many

5 h

2 Vanilla RNN
- Hidden State
- BPTT
- Vanishing Gradients

5 h

3 LSTM
- Input / Forget / Output Gates
- Cell State

5 h

4 GRU
- Update & Reset Gates
- GRU vs LSTM

5 h

5 Seq2Seq
- Encoder–Decoder
- Bahdanau Attention

5 h

6 Variants
- Bidirectional
- Stacked
- 1D CNNs

Recommended 2 h

7 State-Space Models
- Mamba

Optional 1 h

S30 Attention & Transformers

60 h ÷ ≈ 30 days at 2 h/day

1 Attention
- Queries / Keys / Values
- Scaled Dot-Product
- Soft Lookup

7 h

2 Self-Attention
- Multi-Head Attention
- Causal Masking

7 h

3 Positional Encoding
- Sinusoidal
- Learned
- RoPE

7 h

4 Transformer Block
- Attention + MLP
- Residuals
- LayerNorm
- Pre-norm vs Post-norm

7 h

5 Model Families
- Encoder-only (BERT)
- Decoder-only (GPT, Llama)
- Encoder–Decoder (T5)

8 h

6 Build GPT from Scratch
- Char-level Model
- BPE Tokens
- nanoGPT

8 h

7 Complexity
- O(n²) Attention
- Context Length
- KV Cache

8 h

8 Efficient Attention
- FlashAttention
- GQA / MQA
- Sliding Window
- Mixture of Experts

Recommended 4 h

9 Multimodal Transformers
- Vision Transformers
- Image–Text Models

Recommended 4 h

S31 Natural Language Processing

50 h ÷ ≈ 25 days at 2 h/day

1 Text Preprocessing
- Normalization
- Tokenization
- Stemming / Lemmatization
- Stop Words
- Regex

6 h

2 Classical Representations
- Bag of Words
- TF-IDF
- n-gram LMs

6 h

3 Word Embeddings
- word2vec
- GloVe
- FastText
- Analogies

6 h

4 Classical Tasks
- Text Classification
- Sentiment
- NER
- POS Tagging
- spaCy

6 h

5 Transformer NLP
- HF Pipelines
- AutoTokenizer
- AutoModel
- Trainer API
- Fine-tuning BERT

6 h

6 Seq2Seq Tasks
- Summarization
- Translation
- Question Answering

6 h

7 NLP Evaluation
- Accuracy / F1
- BLEU
- ROUGE
- BERTScore

6 h

8 Semantic Search
- Sentence Embeddings

Recommended 3 h

9 Multilingual NLP
- Indian Languages
- IndicBERT

Recommended 3 h

10 Speech
- Whisper ASR
- Text-to-Speech

Optional 2 h

S32 Generative Models

40 h ÷ ≈ 20 days at 2 h/day

1 Autoencoders
- Encoder / Bottleneck / Decoder
- Denoising AE
- Anomaly Detection

7 h

2 VAEs
- Latent Variables
- Reparameterization
- ELBO

7 h

3 GANs
- Generator vs Discriminator
- Minimax
- Mode Collapse
- DCGAN
- StyleGAN

7 h

4 Diffusion Models
- Forward Noising
- DDPM
- Sampling Steps
- Classifier-Free Guidance

7 h

5 Latent Diffusion
- VAE + U-Net / DiT
- Text Encoder
- diffusers Library

7 h

6 Fine-tuning Diffusion
- LoRA
- DreamBooth
- ControlNet

Recommended 3 h

7 Frontier Generative
- Flow Matching
- Video Generation
- Audio Generation

Optional 2 h

Continues: Phase 6 · Generative AI & LLMs

PHASE 6

Generative AI & LLMs

285 hours · ≈ 143 study days at 2 h/day

S33 — LLM Fundamentals, Tokenization & Embeddings

40 h · ≈ 20 days at 2 h/day

1 Tokenization

BPE · WordPiece · SentencePiece · Vocabulary Size · tiktoken · BPE from Scratch

6 h

2 Embeddings

Token vs Sentence Embeddings · Embedding Models · Similarity Metrics · MTEB

5 h

3 Pretraining

Next-Token Prediction · Data Mixtures · Scaling Laws

5 h

4 Post-training

SFT / Instruction Tuning · RLHF · DPO · RL for Reasoning

5 h

5 Decoding

Logits & Softmax · Temperature · Top-k / Top-p · Greedy vs Sampling · Beam Search · KV Cache

5 h

6 Model Landscape

Closed APIs · Open-Weight Models · Model Cards · Licences · Choosing a Model

5 h

7 Limitations

Hallucination · Knowledge Cutoff · Sycophancy · Lost-in-the-Middle

5 h

8 Multimodal LLMs

Vision-Language · Audio Models

Recommended 2 h

9 Reasoning Models

Test-Time Compute · Thinking Budgets

Recommended 2 h

S34 — Prompt Engineering & LLM APIs

30 h · ≈ 15 days at 2 h/day

1 API Basics

Messages & Roles · Parameters · Streaming · Token Counting · Retries & Rate Limits · Cost Tracking

3 h

2 Prompt Techniques

Clear Instructions · Few-Shot · Delimiters / XML · Chain of Thought · Prompt Chaining

3 h

3 Structured Output

JSON Mode · Schemas · Pydantic · Instructor

4 h

4 Tool Use

Function Calling · Tool Schemas · Call–Execute–Return Loop

4 h

5 Prompt Management

Versioning · Templates · Testing on Datasets

4 h

6 Cost & Latency

Prompt Caching · Batch APIs · Smaller Models · Response Caching

4 h

7 Safety Basics

Prompt Injection · Jailbreaks · Output Filtering

4 h

8 Local Models

Ollama · llama.cpp · OpenAI-compatible APIs

Recommended 2 h

9 Prompt Optimization

DSPy

Recommended 2 h

S35 — Fine-tuning & PEFT

50 h · ≈ 25 days at 2 h/day

1 When to Fine-tune

Prompting vs RAG vs Fine-tuning

6 h

2 Full Fine-tuning

HF Trainer · Memory Needs

6 h

3 Fine-tuning Data

Instruction Formats · Chat Templates · Synthetic Data · Train / Eval Splits

6 h

4 PEFT

LoRA · Rank & Alpha · QLoRA · Adapters · Prefix Tuning · Merging Adapters

6 h

5 Tooling

PEFT Library · TRL (SFT / DPO) · Unsloth · Axolotl

6 h

6 Preference Tuning

DPO · Reward Models · PPO · GRPO

6 h

7 Evaluating Fine-tunes

Held-out Sets · LLM-as-Judge · Catastrophic Forgetting

6 h

8 Embedding Fine-tuning

sentence-transformers

Recommended 3 h

9 Distillation

Teacher → Student

Recommended 3 h

10 Advanced

Continued Pretraining · Model Merging

Optional 2 h

S36 — RAG & Vector Databases

60 h · ≈ 30 days at 2 h/day

1 RAG Pipeline

Ingest · Parse · Chunk · Embed · Index · Retrieve · Rerank · Generate

6 h

2 Document Parsing

PDF · HTML · Tables · OCR · Metadata

6 h

3 Chunking

Fixed-size · Recursive · Structural · Semantic · Overlap

7 h

4 Vector Search

Exact vs ANN · Flat · IVF · HNSW · Product Quantization · Recall vs Latency

7 h

5 Vector Databases

FAISS · Chroma · Qdrant · Pinecone · pgvector · Weaviate / Milvus · Metadata Filters

7 h

6 Better Retrieval

Hybrid (BM25 + Dense) · RRF · Rerankers · Query Rewriting · HyDE · Multi-Query

7 h

7 Generation

Context Templates · Citations · "I Don't Know" · Long Context

7 h

8 RAG Evaluation

Recall@k · MRR · Faithfulness · Answer Relevance · Ragas · Golden Sets

7 h

9 Advanced RAG

Agentic RAG · GraphRAG · Multimodal RAG · Parent-Document · Self-RAG

Recommended 3 h

10 RAG Operations

Re-indexing · Access Control · Caching · Cost

Recommended 3 h

S37 — AI Agents & Agentic AI

55 h · ≈ 28 days at 2 h/day

1 Workflows vs Agents

Prompt Chains · Routing · Parallelization · Orchestrator–Workers · Evaluator–Optimizer

5 h

2 Agent Loop

ReAct · Stop Conditions · Max Steps · Error Recovery

5 h

3 Tools

Tool Design · Descriptions · Validation · Safe Tools · Code Sandboxes

6 h

4 Memory & State

Short-Term · Long-Term · Context Management · Summarization

6 h

5 Planning

Task Decomposition · Self-Critique · Retries

6 h

6 Frameworks

Agent from Scratch · LangGraph · OpenAI Agents SDK · Claude Agent SDK · LlamaIndex · CrewAI

6 h

7 MCP

Model Context Protocol · Build an MCP Server

6 h

8 Guardrails

Human-in-the-Loop · Permissions · Audit Logs

6 h

9 Agent Evaluation

Success Rate · Trajectory Review · Cost per Task · SWE-bench

6 h

10 Computer-Use Agents

Browser Agents · Desktop Agents

Recommended 3 h

S38 — Multi-Agent Systems

30 h · ≈ 15 days at 2 h/day

1 Architectures

Supervisor · Hierarchical · Peer-to-Peer · Sequential · Debate / Critic

7 h

2 Communication

Shared State · Message Passing · Handoffs · A2A Protocol

6 h

3 Implementations

LangGraph Multi-Agent · CrewAI · AutoGen

6 h

4 Failure Modes

Loops · Cost Explosions · Conflicts · Error Propagation · Tracing

6 h

5 Single vs Multi

Cost & Latency Comparison

Recommended 3 h

6 Simulations

Generative Agents · Multi-Agent RL

Optional 2 h

S39 — LLM Evaluation

20 h · ≈ 10 days at 2 h/day

1 Error Analysis

Read Outputs · Categorize Failures

4 h

2 Eval Datasets

Golden Sets · Synthetic Tests · Adversarial Cases · No Train Leakage

3 h

3 Metric Types

Rule-based · Reference-based · LLM-as-Judge · Human Eval

3 h

4 Eval Tooling

DeepEval · promptfoo · Ragas · Langfuse / Phoenix

3 h

5 Regression Testing

Eval Gates in CI

3 h

6 Public Benchmarks

MMLU · GSM8K · HumanEval · SWE-bench · Contamination

Recommended 2 h

7 Online Evaluation

User Feedback · A/B Tests

Recommended 2 h

Continues: Phase 7 · Production AI Engineering

PHASE 2

Production AI Engineering

460 hours → 230 study days at 2 h/day

041 Docker & Containerization
40 h → 40 days at 2 h/day

1 D1 Containerization Fundamentals
- Works-on-My-Machine
- VMs vs Containers
- Images vs Containers
- Install Docker / WSL2

3 h

2 D4 Dockerfiles & Optimization
- Instructions
- CMD vs ENTRYPOINT
- .dockerignore
- Build Cache
- BuildKit
- Base Images
- Multi-stage Builds
- Hadolint / dive

4 h

3 D6 Compose, Env & Secrets
- compose.yaml
- depends_on + Healthchecks
- .env Files
- Secrets
- compose watch
- Profiles

3 h

4 D8 Registries & Debugging
- Docker Hub
- GHCR
- ECR / ACR
- Tagging Strategy
- Exit Codes
- Troubleshooting

4 h

5 D10 Docker for ML Development
- Reproducible Envs
- ML Base Images
- Jupyter in Docker
- Dev Containers
- Postgres + pgvector

7 h

6 D12 Production AI & Kubernetes
- Graceful Shutdown
- Health Endpoints
- Full RAG Stack
- Compose → K8s
- Helm
- Pipeline Step Images

13 h

7 Container Registries *Recommended*
- Docker Hub
- GHCR
- ECR
- ACR

1 h

8 Container Debugging *Recommended*
- docker exec
- docker logs
- docker inspect
- docker stats

1 h

9 Docker Networking *Recommended*
- Bridge
- Host
- Overlay
- DNS

1 h

10 Docker Volumes *Recommended*
- Named Volumes
- Bind Mounts
- tmpfs
- Backup Patterns

1 h

11 Docker Security *Recommended*
- Non-root User
- Capabilities
- Read-only FS
- Image Scanning (Trivy)
- Memory / CPU Limits
- OOM (137)

2 h

12 Docker in CI/CD & Cloud *Recommended*
- GitHub Actions Builds
- Layer Caching
- Build Once, Promote
- ECS / Cloud Run

4 h

13 D11 GPU & LLM Inference *Recommended*
- NVIDIA Container Toolkit
- --gpus
- CUDA Tags
- GPUs in Compose
- vLLM / Ollama Containers
- Model Weight Volumes

4 h

042 Kubernetes & Orchestration
65 h → 33 days at 2 h/day

1 Orchestration
- Scheduling
- Self-Healing
- Scaling
- Desired State

3 h

2 Core Objects
- Pods
- ReplicaSets
- Deployments
- Services
- Namespaces
- Labels

3 h

3 Storage
- Volumes
- PV / PVC
- StorageClasses

4 h

4 Networking
- Services
- Ingress / Gateway API
- Cluster DNS

4 h

5 Packaging
- Helm
- Kustomize

4 h

6 GPUs on K8s *Recommended*
- Device Plugin
- GPU Operator
- Taints & Tolerations

3 h

7 Extending K8s *Optional*
- Operators
- CRDs
- Istio

1 h

8 Architecture
- API Server
- etcd
- Scheduler
- Controllers
- kubelet
- kube-proxy

5 h

9 Configuration
- ConfigMaps
- Secrets
- Requests & Limits

5 h

10 kubectl
- apply
- get / describe
- logs / exec
- port-forward
- CrashLoopBackOff

5 h

11 Health & Scaling
- Liveness
- Readiness
- Startup Probes
- HPA

4 h

12 Batch Work
- Jobs
- CronJobs

4 h

13 ML on K8s *Recommended*
- Kserve
- Kubeflow
- KubeRay
- KEDA

3 h

14 GitOps *Recommended*
- Argo CD
- Flux

3 h

043 CI/CD
25 h → 13 days at 2 h/day

1 Concepts
- CI vs CD
- Pipelines
- Artifacts

3 h

2 Python CI
- Lint
- Type-check
- Test
- Coverage

4 h

3 Deployment Strategies
- Rolling
- Blue-Green
- Canary
- Shadow

4 h

4 Infrastructure as Code *Recommended*
- Terraform Basics

2 h

5 GitHub Actions
- Workflows
- Triggers
- Jobs & Steps
- Runners
- Matrix
- Caching
- Secrets

3 h

6 Container CI
- Build
- Scan
- Push
- Tagging

4 h

7 ML CI
- Data Validation
- Model Tests
- Eval Gates
- Registry Promotion

4 h

8 Other CI Systems *Optional*
- GitLab CI
- Jenkins

1 h

044 Cloud Computing (AWS, Azure, GCP)
70 h → 35 days at 2 h/day

1 Cloud Basics
- Regions & AZs
- Shared Responsibility
- Pricing Models
- Free Tier
- Billing Alerts

6 h

2 Storage
- Object Storage
- Block Storage
- Managed Databases

7 h

3 Managed Kubernetes
- EKS
- AKS
- GKE

7 h

4 Managed ML
- SageMaker
- Azure ML
- Vertex AI

7 h

5 Terraform
- Providers
- Resources
- State
- Modules

7 h

6 GPU Clouds *Recommended*
- Lambda
- RunPod
- Modal

3 h

7 Identity & Access
- IAM Users
- Roles
- Policies
- Least Privilege

7 h

8 Compute
- VMs
- GPU Instances
- Serverless Functions
- Container Services

7 h

9 Networking
- VPC
- Subnets
- Security Groups
- Load Balancers

7 h

10 Managed LLMs
- Bedrock
- Azure OpenAI
- Vertex Gemini

7 h

11 Cost Management
- Right-sizing
- Spot
- Scale to Zero
- Budgets

Recommended 3 h

12 Certifications *Optional*
- AWS AI Practitioner
- AWS ML Engineer
- Azure AI Engineer
- GCP ML Engineer

2 h

045 Model Optimization & Quantization
30 h → 15 days at 2 h/day

1 Numeric Formats
- fp32
- fp16
- bf16
- fp8
- int8
- int4

3 h

2 LLM Quantization
- bitsandbytes
- GPTQ
- AWQ
- GGUF Levels
- Quality Checks

5 h

3 Compilers & Runtimes
- torch.compile
- ONNX Runtime
- TensorRT-LLM
- OpenVINO

5 h

4 Edge Deployment *Optional*
- Core ML
- LiteRT
- ExecuTorch

1 h

5 Quantization Basics
- Scale & Zero-point
- Per-tensor vs Per-channel
- PTQ vs QAT

5 h

6 Other Techniques
- Pruning
- Distillation
- Low-rank Factorization

5 h

7 Kernel Optimizations
- FlashAttention
- Fused Kernels

Recommended 2 h

8 Benchmarking
- Warmup
- Realistic Inputs
- Quality + Speed

Recommended 2 h

046 Model Serving & Inference
45 h → 23 days at 2 h/day

1 Serving Patterns
- Online
- Batch
- Streaming
- Edge vs Cloud

4 h

2 Performance Metrics
- p50 / p99 / p99
- Throughput
- TTFT
- Tokens per Second
- Cost per Request

5 h

3 General Servers
- Triton
- TorchServe
- BentoML
- Ray Serve

5 h

4 Scaling
- Horizontal
- GPU Memory Planning
- Autoscaling
- Cold Starts
- Queues

5 h

5 LLM Gateways *Recommended*
- LiteLLM
- Portkey
- Per-user Limits

2 h

6 Simple Serving
- FastAPI Model API
- Load at Startup
- Health Checks

5 h

7 Batching
- Static
- Dynamic
- Continuous Batching

5 h

8 LLM Servers
- vLLM
- SGLang
- TGI
- Ollama / llama.cpp
- NVIDIA NIM

5 h

9 Model Formats
- safetensors
- ONNX
- torch.export
- GGUF
- TensorRT

5 h

10 Advanced LLM Serving
- Prefix Caching
- Speculative Decoding
- Tensor Parallel
- Multi-LoRA
- Routing

Recommended 5 h

047 MLOps & LLMOps
60 h → 30 days at 2 h/day

1 MLOps Maturity
- Level 0
- Level 1
- Level 2

4 h

2 Versioning
- DVC
- HF Hub
- Seeds & Configs
- Environment Locks

7 h

3 ML Pipelines
- Airflow
- Prefect
- Kubeflow Pipelines
- ZenML
- Retraining Triggers

7 h

4 Testing ML Systems
- Data Tests
- Model Tests
- Infra Tests

7 h

5 Governance
- Continuous Training
- Approval Workflows

Recommended 3 h

6 Platforms *Optional*
- Build vs Buy
- Databricks
- SageMaker
- Vertex AI

2 h

7 Experiment Tracking
- MLflow
- Weights & Biases
- Params, Metrics, Artifacts

7 h

8 Model Registry
- Versions
- Stages
- Lineage
- Model Cards

7 h

9 Feature Stores
- Offline / Online
- Point-in-time Correctness
- Feast

7 h

10 LLMOps
- Prompt Versioning
- Eval Pipelines
- Tracing
- Guardrails
- Cost per Feature
- Fallbacks

7 h

048 Monitoring & Observability
25 h → 13 days at 2 h/day

1 Three Pillars
- Metrics
- Logs
- Traces
- Structured Logging
- Correlation IDs

3 h

2 SLOs
- SLI / SLO / SLA
- Error Budgets
- Golden Signals

4 h

3 ML Monitoring
- Data Drift
- Concept Drift
- Prediction Drift
- PSI / KS
- Evidently

4 h

4 Incident Response
- On-call
- Runbooks
- Postmortems

Recommended 2 h

5 Infra Monitoring
- Prometheus
- PromQL
- Grafana
- DCGM GPU Metrics
- Alerting

4 h

6 Tracing
- OpenTelemetry
- Collector
- Spans
- GenAI Conventions

4 h

7 LLM Monitoring
- Tokens & Cost
- Step Latency
- Hallucination Flags
- User Feedback
- Trace Review

4 h

049 AI Security & Responsible AI
25 h → 13 days at 2 h/day

1 LLM Threats
- OWASP LLM Top 10
- Prompt Injection
- Data Leakage
- Insecure Output
- Excessive Agency
- Poisoning

2 h

2 App Security
- Secrets
- Dependency Scanning
- Container Security
- Auth
- Encryption
- Audit Logs

3 h

3 Fairness & Bias
- Sources of Bias
- Fairness Metrics
- Mitigation
- Fairlearn

3 h

4 Governance
- NIST AI RMF
- EU AI Act
- India DPDP Act
- AI Policies

3 h

5 Guardrail Tools *Recommended*
- NeMo Guardrails
- Llama Guard

2 h

6 Defences
- Least-Privilege Tools
- I/O Filtering
- Instruction / Data Separation
- Human Approval
- Sandboxing
- Red-Teaming

3 h

7 Privacy
- PII Redaction
- Data Minimization
- Retention
- Differential Privacy

3 h

8 Transparency
- Model Cards
- Datasheets
- Explainability

3 h

9 Adversarial ML
- Adversarial Examples
- Model Extraction
- Membership Inference
- MITRE ATLAS

Recommended 2 h

10 AI Safety Research *Optional*
- Alignment
- Interpretability

1 h

050 System Design for AI
35 h → 18 days at 2 h/day

1 Design Process
- Requirements
- Scale Estimates
- High-Level Design
- Deep Dives
- Trade-offs

4 h

2 ML Design Patterns
- Offline Train / Online Serve
- Feature Pipelines
- Retrieval + Ranking
- Feedback Loops

5 h

3 Reliability
- Redundancy
- Graceful Degradation
- Timeouts & Retries
- Circuit Breakers

3 h

4 Practice Cases
- Recommender
- Fraud Detection
- Enterprise Search
- Coding Assistant

Recommended 8 h

5 Distributed Building Blocks
- Load Balancers
- Caches
- Queues
- Replication
- Sharding
- Idempotency
- Back-pressure

4 h

6 LLM System Design
- RAG at Scale
- Agent Platforms
- Model Routing
- Caching
- Multi-tenancy
- Cost Budgets

5 h

7 Cost Modelling
- GPU-hours
- Tokens
- API vs Self-hosted

3 h

8 Documentation
- Design Docs
- Architecture Diagrams
- ADRs

3 h

Continues: Phase 3 · Advanced & Research

PHASE 8

Advanced & Research

290 hours · ≈ 145 study days at 2 h/day

S50 Reinforcement Learning

80 h · ≈ 40 days at 2 h/day

1 RL Problem

Agent & Environment | States, Actions, Rewards | Exploration vs Exploitation | Bandits

10 h

2 MDPs

Returns | Discounting | Policies | Value Functions | Bellman Equations

9 h

3 Dynamic Programming

Policy Iteration | Value Iteration

9 h

4 Model-free Methods

Monte Carlo | TD Learning | SARSA | Q-learning

9 h

5 Deep Q-Networks

Experience Replay | Target Networks

9 h

6 Policy Gradients

REINFORCE | Baselines | Actor–Critic | GAE

9 h

7 PPO

Clipped Objective | RLHF Workhorse

9 h

8 RL for LLMs

Reward Models | RLHF Pipeline | DPO | GRPO | Verifiable Rewards

9 h

9 RL Tools

Gymnasium | Stable-Baselines3 | CleanRL | TRL

Recommended 5 h

10 Advanced RL

Model-based | Offline RL | Multi-agent RL | Imitation Learning

Optional 2 h

S51 Distributed Training & GPU Computing

60 h · ≈ 30 days at 2 h/day

1 GPU Architecture

SMs | CUDA vs Tensor Cores | HBM & Shared Memory | Roofline | Arithmetic Intensity

7 h

2 Memory Budget

Weights | Gradients | Optimizer States | Activations | Checkpointing | Grad Accumulation

7 h

3 Mixed Precision

autocast | Loss Scaling | bf16

7 h

4 Profiling

PyTorch Profiler | nvidia-smi | Nsight | Data-loading Bottlenecks

7 h

5 Data Parallelism

DDP | All-reduce | NCCL | torchrun

7 h

6 Sharded Parallelism

FSDP / FSDP2 | DeepSpeed ZeRO 1–3

7 h

7 Model Parallelism

Tensor | Pipeline | Sequence / Context | Expert | 3D Parallelism

Recommended 4 h

8 Frameworks

Accelerate | DeepSpeed | Ray Train | torchtitan

Recommended 4 h

9 GPU Programming

Kernels, Threads, Blocks | Memory Coalescing | Triton | CUDA C++

Recommended 4 h

10 Cluster Ops

Slurm | Multi-node | Fault Tolerance | K8s Training Operators

Recommended 4 h

11 Other Accelerators

TPUs / XLA | AMD ROCm | Trainium

Optional 2 h

S52 Research Track

120 h · ≈ 60 days at 2 h/day

1 Reading Papers

Three-Pass Method | Reading Log

24 h

2 Math to Code

Notation Tables | Shapes as Comments | Tiny Worked Examples

24 h

3 Interpreting Experiments

Baselines | Ablations | Error Bars | Contamination

24 h

4 Reproduction

Run Official Code | Small-scale Reimplementation | Match a Table / Figure

24 h

5 Own Experiments

Hypotheses | Research Gaps | Ablations

Recommended 12 h

6 Writing

Blog Posts | Technical Reports | Workshop Papers

Recommended 12 h

S53 Emerging AI (ongoing)

30 h · ≈ 15 days at 2 h/day

1 Reasoning & Test-Time Compute

Reasoning Models | Verifiers | Inference-time Search

Optional 5 h

2 Agents at Scale

Long-horizon Agents | Agent Protocols | Agent Evaluation

Optional 5 h

3 Efficient Architectures

Mamba / SSMs | Hybrid Models | MoE | Long Context

Optional 4 h

4 Multimodality

Any-to-Any Models | Video Generation | Speech-native Assistants

Optional 4 h

5 Small & On-device Models

Distillation | Edge Inference

Optional 4 h

6 Interpretability & Safety

Sparse Autoencoders | Circuits | Scalable Oversight

Optional 4 h

7 AI for Science & Code

Protein Models | Math Provers | Automated Research

Optional 4 h

Continues: Phase 9 · Capstone & Job Readiness

PHASE 9

Capstone & Job Readiness

160 hours · ≈ 80 study days at 2 h/day

S54 Production Capstones

120 h · ≈ 60 days at 2 h/day

1 Capstone A — LLM Platform

- RAG + Agents
- Multi-tenant Auth
- Model Routing
- Kubernetes
- CI/CD
- Monitoring
- Security

60 h

2 Capstone B — MLOps Pipeline

- Data Validation
- Retraining
- Model Registry
- Canary Deploys
- Drift Alerts
- Rollback

60 h

S55 Interview & Portfolio Prep

40 h · ≈ 20 days at 2 h/day

1 Coding Interviews

- NeetCode 150
- Timed Practice

8 h

2 ML Fundamentals Interviews

- Bias–Variance
- Metrics
- Backprop
- Attention

7 h

3 System Design Interviews

- ML / LLM Design Cases
- Diagrams

7 h

4 Project Deep-Dives

- Design Decisions
- Failures & Fixes

7 h

5 Portfolio

- Pinned Repos
- Demo Videos
- Blog
- LinkedIn

7 h

6 Behavioural Interviews

- STAR Stories

Recommended

4 h

FINAL

Total Duration at 2 Hours a Day

2,755 hours ≈ 63 months ≈ 5.3 years

Phase	Name	Subjects	Hours	Study days	Weeks	Months
Phase 0	Computing Foundations	4	82	41	7	1.9
Phase 1	Programming & Software Engineering	4	265	133	22	6.1
Phase 2	Mathematics for AI (parallel track)	3	215	108	18	4.9
Phase 3	DSA, Data & Systems	7	370	185	31	8.5
Phase 4	Classical Machine Learning	7	285	143	24	6.6
Phase 5	Deep Learning	7	343	172	29	7.9
Phase 6	Generative AI & LLMs	7	285	143	24	6.6
Phase 7	Production AI Engineering	10	460	230	38	10.6
Phase 8	Advanced & Research	4	290	145	24	6.7
Phase 9	Capstone & Job Readiness	2	160	80	13	3.7
Total		**55**	**2,755**	**1,378**	**230**	**63**

How this is counted: 2 hours a day, 6 days a week (12 hours a week). Study days = hours ÷ 2. Weeks = study days ÷ 6. Months include one revision week after every 5 learning weeks.
Shorter route: skipping Phase 8 (Advanced & Research) leaves 2,465 hours ≈ 57 months at 2 h/day, enough for a first AI/ML engineering job.
Faster pace: at 4 h/day the full roadmap takes about half as long; at 6 h/day about a third.